

Unit 9 Seminar

Principles of Artificial Intelligence and Its Applications

1. Icebreaker Activity (10 minutes) – AI in Everyday Life

- ChatGPT
- Visual Code
- Google Search
- Capatchca
- Mobile phone keyboard prediction

How does AI make decisions in your chosen example?

What kind of data does the AI system rely on?

ChatGPT

- Uses a large trained data set and learnt behaviours:
- Most probabilistic next word
- Books
- Images
- Videos
- Reddit
- Published papers
- Etc
- Additional instructions, such as project files
- Predefined limitations/legal requirements -e.g. Can no create fake nudes or hack a programmes key-code

Visual Studio Code

- Probablastic next line of code
- Stackoverflow?
- Other users code?
- Trained data sets?
- Predefined restrictions/rules? Tied to coding language?
- Can use custom coding guidelines per company

Google Search

- Probabilistic based on:
 - Previous behaviours
 - Similar users - local, age, gender, IP address??
 - Sponsorship
- User data
- Legal restrictions - GDPR?

CAPTCHA (Completely Automated Public Turing test to tell Computers and Humans Apart) - most interesting

- Human-powered machine learning algorithm - used to train models, e.g. letter recognition or image recognition - this is a mountain.
<https://www.accessnewswire.com/618585/By-Typing-Captcha-you-are-Actually-Helping-AIs-Training#:~:text=According%20to%20CAPTCHA:%20using%20hard.results%2C%20and%20Google%20Maps%20result.>
- Prevents bots, but not in the way you would think.
- Converts books to text using humans as a “mechanical turk”

2. Introduction to Artificial Intelligence (15 minutes)

Definition of AI – Understanding Narrow AI vs. General AI.

Narrow AI - specific Task, some thing (Like ChatGPT/Claude/Gemmini) feel like general purpose, but aren't technically.

General AI - doesn't actually exist yet, could do multiple tasks and learn new ones.

AGI -

<https://www.tweaktown.com/news/110678/nvidia-ceo-jensen-huang-says-agi-has-been-now-been-achieved/index.html>

Machine Learning vs. Traditional Programming – How AI learns from data rather than following explicit instructions.

Machine learning is trained on data. It recognises patterns, makes decisions and executes. E.g. - give set of image, recognize patterns, for example: all images with red colour have a higher probability of having the word “love” in the media data. Labels future images with the word love.

Traditional programming - Tell a computer to do something e.g. if the image contains colour #DC143C, #8B0000, or #FF6666, label as “love”.

Key AI Techniques – Supervised Learning, Unsupervised Learning, Reinforcement Learning

Supervised learning - human manually classifying images: dog, cat, rabbit, etc

Unsupervised learning - feed 300 books, detect patterns, e.g. grouping horror books

Reinforcement learning - teach the computer the rules of chess, have it play lots of games (even against itself) give “reward” and “penalty” higher positive score.

Activity: Students will categorise real-world AI applications based on these three types of learning.

Supervised	Unsupervised	Reinforcement
Google Photos automatically tagging people in pictures	Netflix recommending movies based on viewing history	Email spam filter deciding if a message is spam or not
Voice recognition in Siri	Bank system grouping customers by spending behavior	AI playing chess against itself to improve
Handwriting recognition (reading digits from images)	Fraud detection system spotting unusual transactions	Robot learning to pick up objects through trial and error
Online store grouping products into categories automatically	News website suggesting articles based on what you read	Self-driving car learning how to park
Language translation system (e.g. Google Translate)	AI clustering similar songs together in Spotify	Game AI learning to play a video game by maximizing score
Detecting tumors in medical scans	Customer segmentation in marketing (grouping similar users)	Chatbot improving responses based on user feedback (thumbs up/down)
	AI organizing a photo library by visual similarity	

	Recommendation system suggesting friends on social media	

Case Study: How Google's AlphaGo defeated human champions in the game of Go using Reinforcement Learning.

2016

<https://blog.google/innovation-and-ai/products/alphago-machine-learning-game-go/>
<https://www.nature.com/articles/nature16961>

Traditional AI methods construct a search tree

2 networks:

- Policy network — selects the next move to play
- Value network — predicts the winner of the game.

Go 10 to the power of 170 possible board configurations

trained the neural networks on 30 million moves from games played by human experts

playing thousands of games between its neural networks Reinforcement learning

Google Cloud Platform

3. Machine Learning and Neural Networks (20 minutes)

How does Machine Learning work?

Data Collection → Data Preprocessing → Model Training → Testing & Deployment.

Types of Neural Networks:

- Feedforward Networks – Used in speech recognition, chatbots.
- Convolutional Neural Networks (CNNs) – Used in image processing and facial recognition.

- Recurrent Neural Networks (RNNs) – Used in natural language processing (NLP).

Activity:

- Students will analyse a simple AI model predicting stock market trends.

They are probabilistic, not definitive, they are limited by their limited training data that may be incomplete, bias or limited.

You can not predict the roll of a die, you can only say that it has a 50% chance of being even or a 1/6 of being a 6.

4. AI Ethics, Bias, and Security Challenges (20 minutes)

Students will explore the ethical and security concerns associated with AI systems.

Meta engineering algorithms that target and addict underage users:

<https://www.bbc.com/news/articles/c747x7qz249o>

Stealing IP - written, visual, graphics, pirated content for training data

<https://www.feltpfairy.com/post/copyright-infringement-in-ai-generated-artwork>

Doesn't share same beliefs as humans - AI does not believe in god/religion.

Elon musk sued for deep fakes -

<https://www.npr.org/2026/03/16/nx-s1-5749490/xai-elon-musk-sexualized-images>

Confidential data may leak into training data

Cybersecurity Risks of AI:

People with little or no prior understanding of hacking can use AI to help them gain a backdoor.

AI in Surveillance use of facial recognition, this somewhat problematic with non-white people.

<https://www.bbc.com/news/technology-33347866#:~:text=Image%20source%2C%20Twitter,dogs%20as%20horses%20in%20May>. - accidentally labeled two black people as gorillas

Confidential data may leak into training data

“Should AI be regulated to prevent bias, job loss, and privacy violations? Why or why not?”

Personally I view AI as a tool, there is no specific ethics around AI compared to any other tool: guns, a hammer, bear trap, compass, calculator,

People should be more aware of the issues around this. However, ultimately the issue is around ethics in general, but not AI in itself.

Should there be more transparency around how the AI made its decisions?

Example: The Amazon AI hiring tool controversy, where AI automatically downgraded female applicants due to biased training data.

Over sampling did not mitigate this in my data oversampling tests.

Interesting topic on Open AI and mental health:

https://www.linkedin.com/posts/last-week-we-brought-together-leaders-from-share-7443003919989035009-neYz?utm_source=share&utm_medium=member_desktop&rcm=ACoAAzAVxsBP00L5wH6uo78KsXYwmvtsGIWDrk